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A Continuous Affinity Ultrafiltration Process for Trypsin Purification

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A continuous process has been devised and tested for purification of a crude trypsin preparation from pig pancreas. The development was based on the principles of affinity chromatography and ultrafiltration. Trypsin was selectively attracted by a water-soluble high molecular weight (>100,000) polymer, bearing a potent and specific trypsin inhibitor, *m*-aminobenzamidine. The trypsin-macroligand complex was then retained by using an appropriate ultrafiltration membrane, while impurities could pass through. The bound trypsin was eluted by either arginine or benzamidine. The process also featured provision for recirculation of the eluant as well as the macroligand. It was demonstrated that this purification process could purify trypsin from the crude preparation with a yield of 77%, contaminated with only 3% of impurities. For the first time, a serious attempt has been made toward continuous purification of enzymes by the affinity ultrafiltration technique. Besides a substantial increase in productivity, the affinity polymer could be easily reconditioned and expected to possess a long operative life. Such characteristics undoubtedly will play an important role in reducing the cost of trypsin purification.

The importance of product purification has been noted by Genentech, the world leading biotechnology company: "Laboratory creation of a new product is the first step. It only begins to count when the product is purified and packaged." The key word of successful commercial biotechnology is "purification" since recovery of a biochemical product always represents a major cost. For instance, the purification of protein products in many recombinant DNA fermentations can contribute up to 90% of the overall processing cost.¹

Extensive processing after fermentation inevitably carries the cost of a large yield loss. Unfortunately, purification of a biochemical product usually requires a series of individual separations, since in many clinical applications products derived from microbial sources must be free from contaminating substances such as pyrogens, native microbial proteins, nucleic acids, etc. In view of this, it is very critical to develop a technique which possesses high product recovery and resolution from impure feed streams.

For the last decade, affinity chromatography, a highly specific technique, has been widely used in industry for the purification of many clinically important products such as interferon, and antithrombin III.² This technique, however, is costly, laborious and time consuming, since the ligand bearing matrices are expensive, difficult to prepare, sus-

ceptible to degradation and only have a limited operating life. Furthermore, the system can only be operated batch-wise and it uses a packed column that can plug and foul. The scaleup of this process is very complicated and the results remain to be seen.

Ultrafiltration, another widely used technique, can be utilized to separate solutes from one another or solutes from solvents. Besides its ease of scaleup, ultrafiltration possesses a high productivity and it can be operated as batch or continuous systems. There are, however, disadvantages to ultrafiltration because of its low resolution. In practice, a resolution of 10-fold difference in molecular weight is about the best that can be expected.

By combining affinity and ultrafiltration, the main purpose of this study was to develop a continuous affinity ultrafiltration process. The system was then used for purifying trypsin from a mixture of trypsin and chymotrypsin and from a porcine pancreatic crude preparation. Following the procedure developed by Luong and co-workers,³ a water-soluble, high-molecular-weight polymer bearing *m*-aminobenzamidine, a strong and specific trypsin inhibitor was synthesized.

This affinity polymer was demonstrated to possess high efficiency in binding trypsin. Consequently, in the presence of this affinity polymer, when a solution of trypsin and chymotrypsin was ultrafiltered with 10⁵ molecular-weight cutoff membranes, 98% of chymotrypsin passed through the membranes, while 90% of trypsin was retained. Furthermore, 90% of the bound trypsin could be released from the binding complex by using arginine as eluant, leaving the polymer regenerated for repeated use. This report presents the experimental results concerning the feasibility of using such an affinity polymer in a continuous process for trypsin purification. Electrophoresis was applied for confirming the purity of the enzymes purified.

MATERIALS AND METHODS

Materials

Trypsin (type III-S), chymotrypsin (type II), crude pig pancreas trypsin (type II), *m*-aminobenzamidine, *N*-benzoyl-L-tyrosine ethyl ester (BTEE), *N*^α-benzoyl-L-arginine ethyl ester (BAEE), *N*^α-benzoyl-L-arginine *p*-nitroanilide (L-BAPNA), *N*-benzoyl-L-tyrosine

p-nitroanilide (L-BTPNA), and dimethyl sulfoxide (DMSO) were purchased from Sigma Chemical Co. (St. Louis, MO). Acrylamide was obtained from Anachemia (Champlain, NY).

Enzyme Assays

Trypsin and chymotrypsin were assayed with 0.5mM BAEE in 50mM Tris, pH 8, 10mM CaCl₂ buffer and with 0.5mM BTEE (dissolved in 50% methanol) in 50mM Tris, pH 8, 50mM CaCl₂ buffer, respectively, at 25°C.⁴

Preparation of the Affinity Polymer

Following the procedure developed by Luong and co-workers,³ *N*-acryloyl-*m*-aminobenzamidine (2 g dissolved in 10 mL DMSO) was added to 350 mL aqueous solution containing 10 g acrylamide. After adding 30 μ L of *N,N,N',N'*-tetramethylethylenediamine (TEMED) and performing extensive flushing of the solution with nitrogen gas to remove oxygen, the polymerization was initiated by adding 40 mL ammonium persulfate solution (concentration of 5 g/L). After 2 h, the resulting solution was diluted four-fold and filtered through the Minitan Millipore ultrafiltration system equipped with 10⁵ cutoff molecular-weight membrane plates. By monitoring the absorbance of the filtrate in the range 190–390 nm, it was observed that more than 80% of the polymer had a molecular weight greater than 10⁵. Measurement of the absorbance at 280 nm also indicated that ca. 90% of *N*-acryloyl-*m*-aminobenzamidine was incorporated into the high-molecular-weight polymeric chains. Only the retentate was used for trypsin purification studies.

Development of a Continuous Affinity-Ultrafiltration Process

A schematic diagram for the continuous affinity ultrafiltration system is shown in Figure 1. Enzyme solution (in

50mM Tris pH 8, 10mM Ca²⁺) containing 100 mg/L each of trypsin and chymotrypsin was peristaltically pumped at a flow rate of 25 mL/min to a container (C1), containing 100 mL affinity polymer (0.75% solution). The feed rate of each of trypsin and chymotrypsin therefore is 150 mg/h. The resulting solution was ultrafiltered through UF1. Pressure was adjusted to set a filtration rate of 25 mL/min resulting in a constant volume in C1. Trypsin was retained (in C1), since it bound to the macromolecular affinity polymer, while unbound chymotrypsin passed through the membranes into the filtrate stream F1. After 2.5 h, fresh polymer (0.75% by weight) was added, concurrently with the enzyme solution, at 0.66 mL/min, to C1 while its content was transferred to C2 at the same rate. After another 2.5 h, the volume accumulated in C2 attained 100 mL, containing mainly the isolated affinity polymer–trypsin complex. At this point, 24 g L-arginine–HCl was added to C2 (yielding 1M arginine solution) and the resulting solution was ultrafiltered by UF2 with a filtration rate of 25 mL/min. Trypsin was dissociated from the affinity polymer and then passed to the filtrate F2 which was collected in C3 (containing 200 mL of 1M L-arginine-HCL in Tris buffer at the beginning). The solution in C3 was processed by UF3 the filtration rate of which was adjusted to 25 mL/min, leaving the volume in C3 constant with trypsin accumulated therein. The filtrate stream F3, containing most of the arginine was recycled to C2, effectively leaving the volume in C2 constant. The retentate of UF2 (remained in C2), mostly made up of affinity polymer in 1M L-arginine–HCl, was transferred at a rate of 0.66 mL/min to C4 containing 100 mL fresh affinity polymer solution (0.75%). Buffer solution was continually added to C4 (ca. 30 mL/min) while its content was processed by UF4 at a filtrate rate of 30 mL/min leaving the volume of C4 constant. Arginine was removed (in F4) thereby leaving the affinity polymer (in C4) regenerated and ready to be recycled to C1. In order to maintain an arginine concentration of 1M in the second ultrafiltration system, 35 mL of 2M L-arginine–HCl solution was added to C2 every hour while 35 mL of F3 was

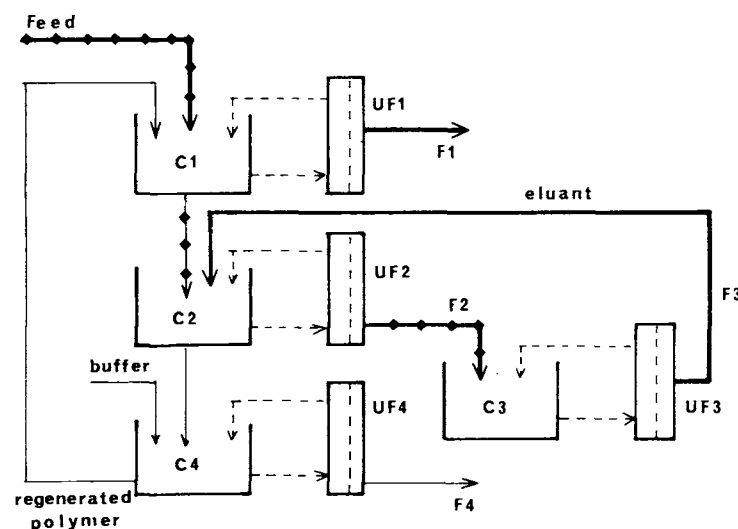


Figure 1. A schematic diagram of the continuous affinity ultrafiltration process.

withdrawn and discarded. Samples from all four containers were withdrawn periodically, the enzyme activities were assayed, and the content of each enzyme in each container was calculated.

In the experiments using the crude preparation, the operation was similar with the following particulars: the crude concentration in the feed was 375 mg/L, total start up time was 10 h, and the eluant was 50mM benzamidine.

All experiments were performed at 4°C. All ultrafiltration units were Millipore Minitan, each was equipped with four membrane plates, each of which had a 60-cm² filtration area. The molecular-weight cutoff of UF3 was 10⁴, other units had a molecular-weight cutoff of 10⁵.

Gel Electrophoresis

The Hoefer Scientific Instruments vertical slab gel electrophoresis unit SE600 was operated at a constant current (60 mA/gel) with the cathode at the bottom, in a continuous buffer system (180mM glycine and 10mM CaCl₂, adjusted to pH 4.0 with acetic acid). In each run, two gels were formed with the same buffer solution containing 11.5% acrylamide, 0.2% bis-acrylamide, and TEMED (225 µL/100 mL). Polymerization was initiated by adding ammonium sulfate (75 mg/100 mL). Pig pancreas crude solution and other samples were loaded into the slots of the slabs. After 4 h operation at 4°C, one gel was stained with Coomassie Brilliant Blue G-250 for protein detection. The second was used for band assignment using an agar overlay technique. The substrate of trypsin or chymotrypsin (L-BAPNA or L-BTPNA, respectively) was added (3mM) to a 1.5% agar solution (in 1M Tris buffer, pH 8.0, containing 10mM CaCl₂). The presence of the enzymes in a band was indicated by the yellow color of the released *p*-nitroanilide.

RESULTS AND DISCUSSION

The preliminary step was to determine whether the contact time between the enzymes and the affinity polymer would affect the binding efficiency. As shown by Luong and co-workers,³ if a mixture of trypsin and chymotrypsin is incubated with the affinity polymer for two hours at 4°C before running the solution through the 10⁵ molecular-weight-cutoff ultrafiltration system, 90% of the trypsin activity remains in the retentate while ca. 10% is detected in the filtrate. At the same time only 2–3% of the chymotrypsin is retained while over 90% of the chymotrypsin activity is observed in the filtrate. Similar results, however, were also obtained if an enzyme solution was added to the polymer and the resulting solution was ultrafiltered immediately. This is an important finding since it implies that one could add the enzyme solution directly to the polymer solution which is being passed through the ultrafiltration unit, instead of having to set up a holding tank for the binding to take place in, prior to running.

Another experiment was conducted to evaluate the system's efficiency to pass chymotrypsin in the presence of

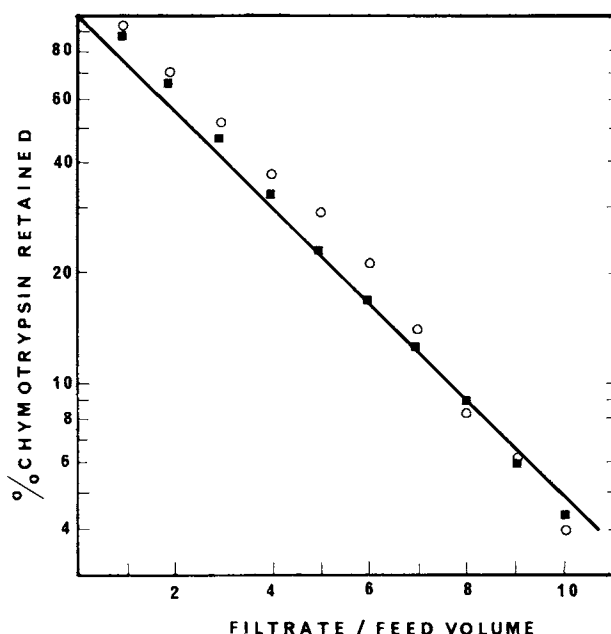


Figure 2. Filtration of chymotrypsin from a 100-mL solution of affinity polymer containing 100 mg each of chymotrypsin and trypsin: (○) filtration rate of 9 mL/min and (■) filtration rate 27 mL/min.

trypsin and the polymer. When the filtration rate was increased threefold (from 9 to 27 mL/min), the same volume ratio still resulted in the same percentage of removal (Fig. 2). Such a finding thus indicates that it is possible to operate at the highest filtration rate (afforded by the system), without sacrificing its effectiveness in passing chymotrypsin. Therefore, a filtration rate of 25 mL/min was used to obtain the results presented hereafter.

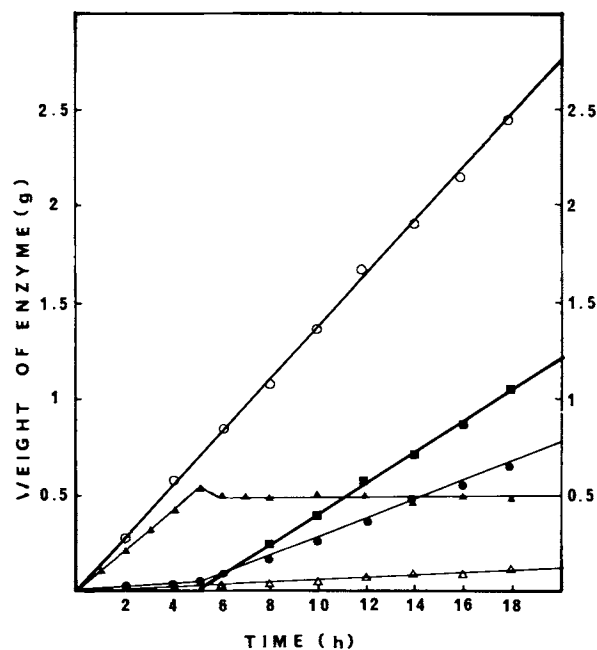


Figure 3. Performance of the continuous system fed with a trypsin-chymotrypsin mixture. Accumulative enzyme contents are presented: (○) chymotrypsin in F1, (●) trypsin in F1, (△) chymotrypsin in holding containers, (▲) trypsin in product container C3.

A series of continuous affinity ultrafiltration experiments was then conducted, a typical set of results is presented in Figure 3. In general, at any time nearly 100% of the introduced chymotrypsin was collected in the filtrate while 4% was detected in the holding containers. About 30% of the starting trypsin was lost in F1 with 60% of the activity being recovered in the four retentates. It should be noted that the enzyme activities (of trypsin and chymotrypsin) detected in the system (by adding activities in various containers) usually do not add up to the input activity due to experimental errors.

As shown in Figure 3, after the startup period, the system quickly attained a steady state as reflected by a constant amount of trypsin in the holding containers. As well, the rate of trypsin accumulation in C3 was almost constant (ca. 80 mg/h) while the weight of chymotrypsin always amounted to 9% that of trypsin. In brief, the continuous process can accomplish a yield of 53.3% at 91% product purity.

The effectiveness of the process was further investigated by using the crude preparation from pig pancreas. This preparation contained 12% trypsin, by weight, and a similar amount of chymotrypsin. With a feed rate of 25 mL/min, at a concentration of 375 mg crude/L, the feed rate of trypsin or chymotrypsin, therefore, is equal to 67.5 mg/h. At the end of the startup period (10 h), a high level of chymotrypsin was detected in C1. This could be due to the fact that the crude preparation contained many other unknown proteins which could bind to the affinity polymer together with trypsin. Such a complex could, in turn, interact with the contaminating chymotrypsin and thus reduce the specificity of the system. Another possibility could be the interaction between chymotrypsin and impurities, thus increasing the apparent molecular weight of chymotrypsin. Although such hypotheses deserve further attention, they are beyond the scope of this report. However, the high level of chymotrypsin in C1 indicated the need for a specific elution procedure. Recourse was therefore attempted through the use of benzamidine, in expectation that only trypsin would be released by this specific inhibitor. This expectation was indeed confirmed by the results presented in Figure 4. At the end of the startup period, 550 mg trypsin was detected in C1, contaminated with 170 mg chymotrypsin (23.6% contamination). However, after 10 h continuous operation, using benzamidine as eluant, the product container C3 exhibited trypsin activity equal to 630 mg contaminated with only 20 mg chymotrypsin (3% contamination). Based on the last 5 h operation, when the rate of trypsin accumulation in C3 was 52 mg/h, the yield is 77% at 97% purity.

Purity of the product was verified by the Coomassie Blue stained electrophoretic gel (Fig. 5). The crude sample resulted in ca. 14 bands (slot 1) while the filtrate (F1) essentially did not exhibit the 2 bands of highest mobility (slot 2). On the contrary, these two bands predominate in slot 3 which was loaded with the final product (taken from container C3). It is therefore clearly indicated that the two high mobility bands correspond to trypsin fractions and that the final product is very slightly contaminated.

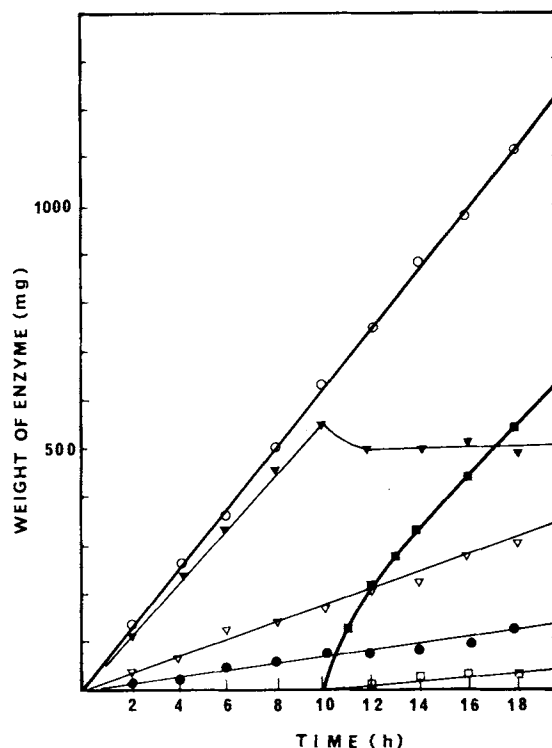


Figure 4. Performance of the continuous system fed with a solution of crude preparation from pig pancreas: (○) chymotrypsin in F1, (●) trypsin in F1, (▽) chymotrypsin in holding containers, (▼) trypsin in holding containers, (■) trypsin in product container C3, and (□) chymotrypsin in C3.

When the electrophoretic gel was overlaid with L-BAPNA (trypsin substrate) containing agar, slot 3 developed strong intensity at the region of the high mobility bands; the color could hardly be observed in slot 2. On the other hand, the L-BTPNA (chymotrypsin substrate) impregnated overlay exhibited strong color development around the middle range bands of slot 2 while no color was visible in slot 3. The assignment of bands as well as purity of the final product were thereby firmly verified.

At first glance, it would appear that the use of benzamidine will incur a high expense. However, the low concentration of benzamidine required (50mM), compared with that of arginine solution (1M) actually would overwhelmingly compensate for the high cost per unit weight of this eluant. Although the purity of trypsin eluted by benzamidine had been verified by the electrophoretic gel, and the yield is very high, the presence in C1 of chymotrypsin and other enzymes is disquieting. Clearly, if these proteins continually accumulated while the system was operated continuously the effectiveness would be adversely affected. In the duration of the experiments conducted in this study the effect did not manifest noticeably. However, some measure will certainly be necessary in a more prolonged procedure. It is envisioned that a solution could be found either with an additional elution step or with a makeup of fresh polymer solution. The choice between the two possibilities would be dictated by the economic analysis.

The demonstrated high resolution and recovery possible with affinity ultrafiltration have given a large impetus to developing its application further. In the last decade, ultra-

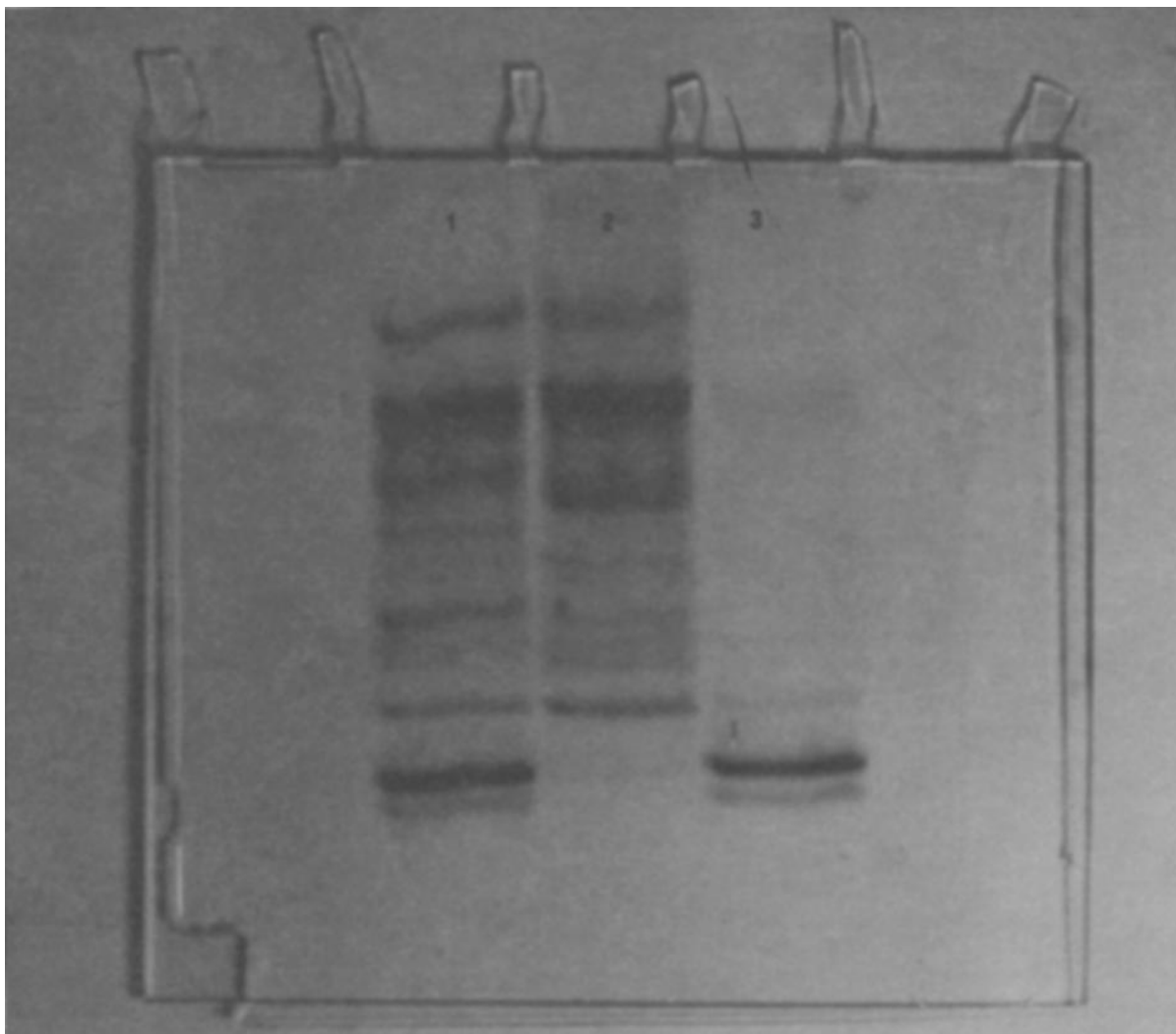


Figure 5. Coomassie Blue stained electrophoretic gel.

filtration has become established as a useful technique for the pharmaceutical industries. This technique has allowed economical and efficient concentration of biochemicals and simple scaleup of purification processes. Nowadays, the membranes are produced in a wide variety of polymers which are resistant to acids, bases, alcohols and temperature, allowing very effective membrane cleaning. Possibilities also exist for fabricating membranes with high-molecular-weight cutoffs between 10^5 and 10^6 , a region of great interest for protein separation. In comparison to conventional affinity chromatography, the affinity ultrafiltration process offers several advantages. The soluble macroligands require no spacer molecules which are often mandatory with insoluble matrices. The synthesis of the macroligand is thus fairly simple, resulting in a low cost, especially when a chemical ligand is used. The soluble ligand is unlikely vulnerable to degradation by attrition or compression, a serious problem which has been experienced in conventional chromatography, where affinity matrices are tightly packed into a column. Last but not least, an instantaneous binding between the ligand and the enzyme indicates a high productivity system.

On the basis of the macroligand stability and the tangential flow of ultrafiltration, it is quite logical to expect that affinity ultrafiltration should have a capability of processing unclarified and viscous broths. The process can also be extended to accommodate solutions containing both soluble and insoluble high-molecular-weight contaminants. In this case, such solutions are preultrafiltered by using a suitable molecular-weight cutoff membrane. This is a well established membrane technology which requires no further development. Undoubtedly, with further refinements the affinity ultrafiltration process would play an important role in biological products purification.

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